Stochastic Optimization
Sample Average Approximation

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Framework

Consider the stochastic program

$$\min_{z \in S} g(z) = \mathbb{E}_P \left[G(z, \xi) \right],$$

- $S \subset \mathbb{R}^m$, compact;
- ξ is a random vector defined on $(\Xi, \mathcal{F}, P)$ and taking values in $(\mathbb{R}^k, \mathcal{B}^k)$
- $G : \mathbb{R}^m \times \mathbb{R}^k \to \mathbb{R}$;
- $\mathbb{E}_P[\cdot]$ is the expectation with respect to P.

We assume that for each $z \in S$, the expected value function $g(z)$ is well defined, i.e. the function $G(z, \cdot)$ is $\mathcal{F}$-measurable and P-integrable.

For simplicity, we assume for now that S is deterministic.

Framework (cont'd)

If ξ cdf is continuous or discrete with a large number of possible realizations, $g(z)$ is usually very difficult to evaluate.

We have to use some approximations such as the Monte Carlo (MC) methods: the original problem is replaced by successive approximations obtained by drawing $\xi_1, \ldots, \xi_n$:

$$\min_{z \in S} \hat{g}_n(z) = \frac{1}{n} \sum_{i=1}^{n} G(z, \xi_i).$$

We sometimes refer to the original program as the true program, or expected value program, and the approximate program as the sample average approximation (SAA).

Two-stage stochastic programming

This framework includes the two-stage programs that we have analyzed, as

$$\min_x \; c^T x + \mathbb{E}[\mathcal{Q}(x)]$$
$$\text{s.t. } Ax = b,\; x \geq 0$$

is equivalent to

$$\min_{x \in S} \mathbb{E}[c^T x + \mathcal{Q}(x)],$$

where $S = \{x \mid Ax = b, x \geq 0\}$

First-order convergence

What happens when $n \to \infty$?

Let z_n^* be a first-order critical point for the approximate problem, i.e. the KKT conditions are satisfied at z_n^*.

$(\xi_1, \ldots, \xi_n)$ can be seen as the finite truncation of an infinite sequence $\overline{\xi} := \{\xi_k\}_{k=1}^{\infty}$.

Solution for a sample size n: $z_n^*(\xi_1, \ldots, \xi_n)$, or $z_n^*\left(\overline{\xi}\right)$.

As S is a compact set, the sequence of SAA solutions has a non-empty set of (finite) limit points.

Under which conditions is such a limit point a first-order critical point for the true problem?

Formalization

Consider the stochastic process

$$\overline{\xi} = \{\xi_k\}_{k=1}^{\infty},$$

later called the *sampling process*, where the random vectors ξ_k, $k = 1, \ldots, \infty$, are assumed to be independent and identically distributed (i.i.d.).

Probability space of infinite dimension

From the i.i.d. property and the Kolmogorov consistency theorem (see for instance Parthasarathy, *Probability Measures on Metric Spaces*, Academic Press, 1967, Chapter V, Theorem 5.1), we can build the probability space of infinite dimension

$$(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi),$$

where the measure P_Π has the property that for any non-zero natural j,

$$P_\Pi[B] = \prod_{i=1}^{j} P[B_i],$$

for any set $B = \prod_{i=1}^{j} B_i \times \prod_{i=j+1}^{\infty} \Xi$, with $B_i \in \mathcal{F}$, $i = 1, \ldots, j$.

Probability space of infinite dimension (cont'd)

In other words, the marginal measures defined on $\prod_{i=1}^{j}(\Xi, \mathcal{F})$, with finite j, $j \geq 1$, correspond to the product measures $\prod_{i=1}^{j} P$, as expected.

We therefore can see $\overline{\xi}$ as a random variable on $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$, whose realizations are processes formed by the successive draws ξ_k, $k = 1, \ldots, \infty$, i.e. $\overline{\xi} = \{\xi_k\}_{k=1}^{\infty}$.

Notations:

- $\xrightarrow{a.s.}$: almost sure;
- $\xrightarrow{p}$: convergence in probability;
- $\Rightarrow$: convergence in distribution.

Probability notions: types of convergence

Definition (Almost sure convergence (or with probability one))

A sequence of random variables $\{\xi_n\}_{n=1}^{+\infty}$ *converges almost surely to a random variable* ξ *if*

$$P_\xi[\{\omega | \, \xi_n(\omega) \to \xi(\omega) \text{ as } n \to \infty\}] = 1.$$

A weaker convergence mode is defined below.

Definition (Convergence in probability)

The sequence of random variables $\{\xi_n\}_{n=1}^{+\infty}$ *converges in probability to the random variable* ξ *if, for all* $\epsilon > 0$,

$$\lim_{n\to\infty} P_\xi[\{\omega \, | \, |\xi_n(\omega) - \xi(\omega)| \geq \epsilon\}] = 0.$$

Probability notions: types of convergence (cont'd)

Almost sure convergence implies converge in probability; the converse is not true.

Definition (Convergence in distribution)

Given a sequence of random variables $\{\xi_n\}_{n=1}^{\infty}$ and a random variable ξ, ξ_n converges in distribution to ξ if ξ_n if

$$\lim_{n\to\infty} F_n(x) \to F(x)$$

for each $x \in \mathbb{R}$ where F is continuous. F_n and F are the distribution functions of ξ_n and ξ, respectively.

We will also note $\xi_n \Rightarrow \xi$.

If both ξ and ζ follow the distribution specified by F, then $\xi_n \Rightarrow \xi$ and $\xi_n \Rightarrow \zeta$ are equivalent statements.

Notations

We will assume that the expressions *almost everywhere* (a.e.) and *almost surely* (a.s.) refer to the infinite dimension space $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$, allowing to consider sets of realizations of the form $\{\xi_n\}_{n=1}^{\infty}$.

Assumptions

A.0 The random draws $\{\xi_k\}_{k=1}^{\infty}$ are i.i.d.

A.1 For P-almost all ξ, the function $G(\cdot, \xi)$ is continuously differentiable on S.

A.2 The family $G(z, \xi)$, $z \in S$, is dominated by a P-integrable function $K(\xi)$, i.e. $\mathbb{E}_P[K]$ is finite and $|G(z, \xi)| \leq K(\boldsymbol{\xi})$ for all $z \in S$ and P-almost each ξ.

Do we need **A.2**?

A.2 is not implied by **A.1**.

Consider ξ s.t.

$$P[\xi = n] = \frac{C}{n^5}, \ n = 1, 2, 3, \ldots,$$

where

$$C = \frac{1}{\sum_{n=1}^{\infty} n^{-5}} = \frac{1}{\zeta(5)} \approx 0.9644$$

We have

$$\mathbb{E}[\xi^2] = C \sum_{n=1}^{\infty} n^2 \frac{1}{n^5} = C \sum_{n=1}^{\infty} \frac{1}{n^3} = \frac{\zeta(3)}{\zeta(5)} < \infty$$

A.1 but not **A.2**

Let $S = [0, 1]$ and

$$G(z, \xi) := \xi^4 z.$$

Then

$$G(z, n) = n^4 z \in C^1 \text{ on } S.$$

So **A.1** holds. Moreover

$$\sup_{z \in S} |G(z, \xi)| = \sup_{z \in S} \xi^4 z = \xi^4.$$

Hence any candidate $K(\xi)$ must satisfy $K(\xi) \geq \xi^4$. Thus

$$\mathbb{E}[K(\xi)] \geq \mathbb{E}\left[\xi^4\right] = C \sum_{n=1}^{\infty} n^4 \frac{1}{n^5} = C \sum_{n=1}^{\infty} \frac{1}{n} = +\infty.$$

Assumptions: discussion

A.1 obviously implies that $G(\cdot, \xi)$ is P-almost surely continuous.

This last property and **A.2**, are typical assumptions in stochastic programming. The stronger form **A.1** is justified by our interest in the first-order optimality conditions.

Uniform law of large numbers

A.0–**A.2** together imply that there exists an uniform law of large numbers (ULLN) on S, for the approximation $\hat{g}_n(z)$ of $g(z)$:

$$\sup_{z \in S} |\hat{g}_n(z) - g(z)| \xrightarrow{a.s.} 0 \quad \text{as } n \to \infty.$$

This also implies that $g(z)$ is continuous on S.

The law of large numbers, its weak and strong forms, and the modes of convergence they use are recalled in the probability background notes. The ULLN strengthens it from pointwise to uniform convergence over S, which is what optimization needs.

Proof: see Theorem 7.48 in Lecture notes on stochastic programming.

Derivatives

Our motivation to study the first-order conditions leads us to introduce an additional assumption on the gradient.

A.3 Each gradient component $\frac{\partial}{\partial [z]_l} G(z, \xi)$ $(l = 1, \ldots, m)$, $z \in S$, is dominated by a P-integrable function.

- **A.3** implies that $\mathbb{E}_P[g(z)]$ is continuously differentiable on S, and $\nabla_z g(z) = \mathbb{E}_P [\nabla_z G(z, \xi)]$.
- Thus, $\mathbb{E}_P[\nabla \hat{g}_n(z^*)] = \nabla g(z^*)$ (unbiased estimator).

First-order convergence

The first-order convergence can be derived from the stochastic variational inequalities:

- A. Shapiro, *Monte Carlo sampling methods*, in A. Shapiro and A. Ruszczyński, editors, *Stochastic Programming*, volume 10 of *Handbooks in Operations Research and Management Science*, pages 353–425. Elsevier, 2003;
- A. Shapiro, *Statistical Inference*, Chapter 5 in *Lectures Notes in Stochastic Programming* (3rd edition), SIAM, 2021.

Stochastic variational inequalities

Consider a mapping $\Phi : \mathbb{R}^m \times \mathbb{R}^k \to \mathbb{R}^m$ and a multi-function $\Gamma : \mathbb{R}^m \rightrightarrows \mathbb{R}^m$, and suppose that $\phi(z) := \mathbb{E}_P[\Phi(z, \xi)]$ is well defined. We denote

$$\phi(z) \in \Gamma(z) \tag{1}$$

as the true generalized equation, or in expected value. $z^* \in \mathbb{R}^m$ is a solution of the generalized equation if $\phi(z^*) \in \Gamma(z^*)$.

If $\{\xi_1, \ldots, \xi_n\}$ is a random sample, we denote

$$\hat{\phi}_n(z) = \frac{1}{n} \sum_{i=1}^{n} \Phi(z, \xi_i) \in \Gamma(z) \tag{2}$$

as the generalized SAA equation, and

- S^*: set of (all) solutions of (1);
- S_n^*: set of (all) solutions of (2).

First-order convergence

Denote by $d(x, A) := \inf_{x' \in A} \|x - x'\|$, the distance from $x \in \mathbb{R}^m$ to A, and $D(A, B) := \sup_{x \in A} D(x, B)$, the deviation of the set A from the set B. We have the following result (Shapiro, 2003, 2022).

Theorem (Generalized equations)

Let $\mathcal{S}$ be a compact subset of $\mathbb{R}^m$ such that $S^* \subseteq \mathcal{S}$. Assume that

(a) *the multi-function $\Gamma(z)$ is closed, i.e. if $z_k \to z$, $y_k \in \Gamma(z_k)$ and $y_k \to y$, then $y \in \Gamma(z)$,*

(b) *the mapping $\phi(z)$ is continuous on $\mathcal{S}$,*

(c) *almost surely, $\emptyset \neq S_n^* \subseteq \mathcal{S}$ for n large enough, and*

(d) *$\hat{\phi}_n(z)$ converges to $\phi(z)$ almost surely uniformly on $\mathcal{S}$ as $n \to \infty$.*

Then $D(S_n^*, S^*) \to 0$ almost surely as $n \to \infty$.

First-order optimality conditions

We denote the feasible set of a mathematical program by $\mathcal{A}$. We will suppose here that $\mathcal{A}$ is closed.

Necessary condition: if x^* is a local minimizer,

$$-\nabla_x f(x^*) \in \mathcal{N}_{\mathcal{A}}(x^*),$$

where $\mathcal{N}_{\mathcal{A}}(x^*)$ is the normal cone to $\mathcal{A}$ at x^*.

Cone

A subset C of a vector space V is a (linear) cone iff αx belongs to C for any $x \in C$ and any strictly positive scalar α; allowing $\alpha = 0$ adds the apex, $0 \in C$.
A convex cone is a cone closed under convex combinations, i.e. iff $\alpha x + \beta y \in C$, $\forall \alpha,\ \beta > 0$, with $\alpha + \beta = 1$.

Not to be confused with a pointed cone, one containing no line, $C \cap -C = \{0\}$ — the property giving a polyhedron its extreme points, and the sense the background notes use.

Tangent cone, normal cone: general framework

We first say that a vector $w \in \mathbb{R}^n$ is tangent to $\mathcal{A}$ at $x \in \mathcal{A}$ if for all sequences of vectors $\{x_i\}$ with $x_i \to x$, and $x_i \in \mathcal{A}$, and all sequences of positive scalars $t_i \downarrow 0$, there exists a sequence $w_i \to w$ such that $x_i + t_i w_i \in \mathcal{A}$ for all i.

The tangent cone $T_{\mathcal{A}}(x)$ is the collection of vectors tangent to $\mathcal{A}$ at x.
The normal cone $N_{\mathcal{A}}(x)$ is its polar cone (background notes), i.e.

$$N_{\mathcal{A}}(x) = \{v \mid v^T w \leq 0,\ \forall w \in T_{\mathcal{A}}(x)\}.$$

The definition above is that of the Clarke tangent cone; the Bouligand one requires only that *some* such sequences exist. The two coincide when $\mathcal{A}$ is convex, which is the only case used below.

OK, but in practice???

Tangent cone, normal cone (cont'd)

If $\mathcal{A}$ is convex, $\mathcal{N}_{\mathcal{A}}(x) = \left\{ v \mid v^T(x - x_0) \geq 0,\ \forall x_0 \in \mathcal{A} \right\}$.

The condition becomes $\nabla_x f(\hat{x})^T(x - \hat{x}) \geq 0$, $\forall x \in \mathcal{A}$.

If f is convex (i.e. we are in the convex programming framework), the condition is also sufficient.

Deterministic and convex constraints

- Thus, when S is convex, we can rewrite the first-order criticality conditions at z^* as
$$-\nabla_z g(z^*) \in \mathcal{N}_S(z^*).$$
- If moreover S is deterministic, the feasible sets are the same for the true and SAA problems.
- Consider $\Gamma(\cdot) = \mathcal{N}_S(\cdot)$; $\phi(z^*)$ belongs to $\Gamma(z^*)$ iff
$$\langle \phi(z^*), u - z^* \rangle \leq 0, \ \forall u \in S.$$
- We refer to such variational inequalities as stochastic variational inequalities.

Convergence

Take $\Phi(z, \xi) = -\nabla_z G(z, \xi)$ and represent by S^* and S_n^* the set of first-order critical points of the true and SAA generalized equations, respectively.

- The assumption (a) holds by definition of the normal cone.
- Then, under **A.0**–**A.3**, we have that $\phi(z) = -\nabla_z g(z)$, and that $\phi(z)$ is a continuous random vector on S, giving assumption (b).
- Assumption (d) follows from the ULLN, while **A.1** and the compactness of S ensure assumption (c) by setting $\mathcal{S} = S$.

The previous theorem therefore ensures the first-order criticality at the limit as $n \to \infty$, almost surely.

Remark

The theorem establishes that $D(S_n^*, S^*) \to 0$ when $n \to \infty$ a.s.
However, we cannot write $D(S^*, S_n^*) \to 0$ when $n \to \infty$ a.s.

Consider the problem

$$\min_{x \in S} |\mathbb{E}[\xi]| x^2,$$

where $S = [-a, a]$, $a > 0$. If $\xi \sim N(0, \sigma^2)$, $S^* = S$. The SAA is

$$\min_{x \in S} \left| \frac{1}{n} \sum_{i=1}^{n} \xi_i \right| x^2,$$

and almost surely, $S_n^* = \{0\}$.
Therefore, a.s. $\lim_{n \to \infty} S_n^* = \{0\} \neq S^*$, and $D(S^*, \{0\}) = a$.

Stochastic constraints

Under stronger assumptions, it is also possible to prove the first-order convergence almost surely when S is non-convex or non-deterministic.

We now assume that S can be described by equality and inequality constraints. The original problem becomes

$$\begin{aligned} \min_{z \in V} \; & g(z) = \mathbb{E}_P[G(z, \xi)], \\ \text{such that } & c_j(z) \geq 0, \; j = 1, \ldots, k, \\ & c_j(z) = 0, \; j = k+1, \ldots, M, \end{aligned}$$

where V is a compact subset of $\mathbb{R}^m$.

Stochastic constraints (cont'd)

SAA problem:

$$\begin{aligned} &\min_{z \in V} \hat{g}_n(z), \\ \text{such that } &\hat{c}_{jn}(z) \geq 0,\ j = 1, \ldots, k, \\ &\hat{c}_{jn}(z) = 0,\ j = k+1, \ldots, M. \end{aligned}$$

Here, for each $j = 1, \ldots, M$, $\{\hat{c}_{jn}(\cdot)\}$ is a sequence of (random) functions converging asymptotically to the corresponding function $c_j(\cdot)$ as $n \to \infty$.

We assume that the functions $c_j(\cdot)$ can be represented in form of expectations:

$$c_j(z) = \mathbb{E}_P[H_j(z, \xi)],\ j = 1, \ldots, M.$$

Stochastic constraints (cont'd)

SAA of the constraints:

$$\hat{c}_{jn}(z) = \frac{1}{n}\sum_{i=1}^{n} H_j(z, \xi_i).$$

For simplicity, we consider the more general parametric mathematical problem

$$\begin{aligned} &\min_{z \in V} \hat{g}(z, \epsilon), \\ \text{such that } &\hat{c}_j(z, \epsilon) \geq 0, \ j = 1, \ldots, k, \\ &\hat{c}_j(z, \epsilon) = 0, \ j = k+1, \ldots, M, \end{aligned}$$

where ϵ is a parametric random vector giving the perturbations of the approximated program, and $g(\cdot)$, $\hat{g}(\cdot, \epsilon)$, $c_j(\cdot)$, $\hat{c}_j(\cdot, \epsilon)$ are assumed to be of class C^2 with respect to z.

Random perturbations

We assume that the perturbation has the form

$$\epsilon = \epsilon(z, \bar{\xi}) = \left(\epsilon_g,\ \epsilon_{c_1},\ \ldots,\ \epsilon_{c_M},\ \epsilon_{\nabla g}^T,\ \epsilon_{\nabla c_1}^T,\ \ldots,\ \epsilon_{\nabla c_M}^T\right)^T,$$

where each component is function from $\mathbb{R}^m \times \prod_{i=1}^{\infty} \mathbb{R}^k$ to $\mathbb{R}$ or $\mathbb{R}^m$, and

$$\hat{g}(z, \epsilon) = g(z) + \epsilon_g,\ \nabla_z \hat{g}(z, \epsilon) = \nabla_z g(z) + \epsilon_{\nabla g},$$
$$\hat{c}_j(z, \epsilon) = c_j(z) + \epsilon_{c_j},\ \nabla_z \hat{c}_j(z, \epsilon) = \nabla_z c_j(z) + \epsilon_{\nabla c_j},\ j = 1, \ldots, M.$$

We also define $\epsilon_n(z, \bar{\xi})$ as

$$\epsilon_n(z, \bar{\xi}) = \begin{pmatrix} \hat{g}_n(z) - g(z) \\ \hat{c}_{jn}(z) - c_j(z),\ j = 1, \ldots, M \\ \nabla_z \hat{g}_n(z) - \nabla_z g(z) \\ \nabla_z \hat{c}_{jn}(z) - \nabla_z c_j(z),\ j = 1, \ldots, M \end{pmatrix},$$

Random perturbations (cont'd)

We denote the corresponding random vector by $\epsilon_n(z, \overline{\xi})$, and we will assume the ULLN holds for the objective and the constraints, as well as for their corresponding derivatives.

We finally assume that the feasible set for the original and approximated problems are non-empty. The Lagrangians of the true and approximated problems are respectively

$$\mathcal{L}(z, \lambda) = g(z) - \sum_{j=1}^{M} [\lambda]_j c_j(z)$$

and

$$L(z, \lambda, \epsilon) = \hat{g}(z, \epsilon) - \sum_{j=1}^{M} [\lambda]_j \hat{c}_j(z, \epsilon).$$

KKT conditions

Let $z^*(\epsilon)$ be a first-order critical point of the approximated program, and assume that $z^*(\epsilon)$ belongs to the interior of V, that we denote by $\overset{o}{V}$.

There exist Lagrange multipliers $\lambda^*(\epsilon)$ such that $(z^*(\epsilon), \lambda^*(\epsilon))$ satisfies the KKT conditions; in other terms $(z^*(\epsilon), \lambda^*(\epsilon))$ is solution to the system

$$
\begin{aligned}
\nabla_z L(z, \lambda, \epsilon) &= 0, \\
[\lambda]_j \hat{c}_j(z, \epsilon) &= 0, \quad j = 1, \dots, M, \\
\hat{c}_j(z, \epsilon) &= 0, \quad j = k+1, \dots, M, \\
\hat{c}_j(z, \epsilon) &\geq 0, \quad j = 1, \dots, k, \\
[\lambda]_j(\epsilon) &\geq 0, \quad j = 1, \dots, k.
\end{aligned}
$$

Solution characteristics

Consider now the specific sampling process $\overline{\xi}$. In order to emphasize the dependency of the first-order critical points to the sampling process, we denote $z_n^*(\overline{\xi})$ for $z^*(\epsilon_n)$ and $\lambda_n^*(\overline{\xi})$ for $\lambda^*(\epsilon_n)$.

Let $\mathcal{Z}\left(\{z_n^*(\overline{\xi})\}\right)$ the set of accumulation points of the sequence $\{z_n^*(\overline{\xi})\}_{n=1}^{\infty}$. The compactness of V implies that for each $\overline{\xi}$, $\mathcal{Z}\left(\{z_n^*(\overline{\xi})\}\right)$ is non-empty.

We can now prove the first-order convergence for the general case.

First-order convergence

Theorem

Assume that for almost every $\overline{\zeta}$ *in* $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$, $\epsilon_n(z, \overline{\zeta}) \to 0$ *uniformly on the compact set* V, *as* $n \to \infty$, *and let* $\overline{\xi}$ *in* $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$ *satisfying the assumption of uniform convergence. If* $z^* \in \mathcal{Z}\left(\{z_n^*(\overline{\xi})\}\right)$ *belongs to* $\overset{o}{V}$, *and a sub-sequence* $\{z_\ell^*(\overline{\xi})\}_{\ell=1}^{\infty} \subseteq \{z_n^*(\overline{\xi})\}_{n=1}^{\infty}$ *converging to* z^* *is associated to a sequence of Lagrange multipliers* $\{\lambda_\ell^*(\overline{\xi})\}_{\ell=1}^{\infty}$ *with a least one limit point, then* z^* *is a first-order critical point of the true problem.*

Proof.

From our assumptions, the sequence $\left\{(z_\ell^*(\overline{\xi}), \lambda_\ell^*(\overline{\xi})\right\}$, $\ell = 1, \ldots, \infty$, has a limit point (z^*, λ^*). The uniform convergence property implied that (z^*, λ^*) satisfies the KKT conditions of the true problem, so that $z^*(\in \overset{o}{V})$ is first-order critical point of the original problem. □

On the Lagrange multipliers

- If $\exists M > 0$ s.t. $\|\lambda_N^*(\bar{\xi})\|$ for all N, $\{\lambda_\ell^*(\bar{\xi})\}_{\ell=1}^{\infty}$ admits a limit point.
- This also allows the use of stochastic variational inequalities (Shapiro, 2003).

Stochastic variational inequalities for KKT

Let $u := (z, \lambda) \in \mathbb{R}^{m+M}$ and $\mathcal{K} := \mathbb{R}^m \times \mathbb{R}^k_+ \times \mathbb{R}^{M-k} \subset \mathbb{R}^{m+M}$. Define

$$\phi(u) = (\nabla_z \mathcal{L}(z, \lambda), c_1(z), \ldots, c_M(z)),$$
$$\hat{\phi}_n(u) = (\nabla_z L(z, \lambda, \epsilon_n), \hat{c}_1(z, \epsilon_n), \ldots, \hat{c}_M(z, \epsilon_n)).$$

The generalized equation $\phi(u) \in \mathcal{N}_{\mathcal{K}}(u)$ then represents the KKT optimality conditions for the original optimization problem, $S^* \subseteq \overset{o}{V}$, and the theorem on generalized equations then implies first-order criticality almost surely, with $\Gamma(u) := \mathcal{N}_{\mathcal{K}}(u)$.
Assumptions (a) and (d) of the theorem on generalized equations are sufficient as $\epsilon \to 0$ a.s., and assumption (c) holds as feasible sets for original and approximate problems are non-empty.

Second-order convergence

If we strengthen our assumptions, we can also show that, almost surely, there exists limit points in $\mathcal{Z}\left(\{z_n^*(\bar{\boldsymbol{\xi}})\}\right)$ that are local minimizers.

We first consider the case where S is deterministic and assume that, for a given sampling process $\bar{\xi}$, $z_n^*(\bar{\xi})$ is a local minimizer of $\hat{g}_n(z)$ over S. In other words,

$$\exists\, \delta_n(\bar{\xi}) \text{ s.t. } \forall z \in B\left(z_n^*(\bar{\xi}), \delta_n(\bar{\xi})\right) \cap S,\ \hat{g}_n\left(z_n^*(\bar{\xi})\right) \leq \hat{g}_n(z).$$

A counter-example

Consider the problem

$$\min_{z\in[-1,1]} z^3 - \frac{z}{2}\mathbb{E}_P[\xi], \tag{3}$$

where $\Xi = \{-1, 1\}$ and $P[\xi = -1] = P[\xi = 1] = 0.5$, so $\mathbb{E}_P[\xi] = 0$, and (3) has one local minimizer only, that is also global, at $z^* = -1$. The SAA problem is then

$$\min_{z\in[-1,1]} z^3 - \frac{z}{2n}\sum_{i=1}^{n} \xi_i. \tag{4}$$

Problem (4) has two (isolated) local minimizers

$$\left\{-1, \sqrt{\frac{\sum_{i=1}^n \xi_i}{6n}}\right\},$$

when $\sum_{i=1}^n \xi_i > 0$. As $n \to \infty$, we have $P\left[\sum_{i=1}^n \xi_i > 0\right] \to 0.5$, but from the ULLN, $\frac{1}{6n}\sum_{i=1}^n \xi_i \to \frac{1}{6}\mathbb{E}_P[\xi] = 0$ a.s.

Rigidity assumption

(Bastin, Cirillo, and Toint, 2006, Pasupathy, 2010)
A.4 There exists some subsequence $\left\{z_\ell^*(\bar{\xi})\right\}_{\ell=1}^{\infty}$ converging to some z^*, with $\left\{z_\ell^*(\bar{\xi})\right\}_{\ell=1}^{\infty} \subseteq \left\{z_N^*(\bar{\xi})\right\}_{N=1}^{\infty}$, and some constants $\delta_{z^*\bar{\xi}} > 0$ and $\ell_{z^*\bar{\xi}} > 0$ such that for all $\ell \geq \ell_{z^*\bar{\xi}}$,

$$\forall z \in B\left(z_\ell^*(\bar{\xi}), \delta_{z^*\bar{\xi}}\right) \cap S,\ \hat{g}_\ell\left(z_\ell^*(\bar{\xi})\right) \leq \hat{g}_\ell(z).$$

Idea: given $\{z_\ell^*(\bar{\xi})\}_{\ell=1}^{\infty} \subseteq \{z_n^*(\bar{\xi})\}_{n=1}^{\infty}$ s.t. $\{z_\ell^* \to z^*$, the neighborhood where $z_\ell^*(\bar{\xi})$ is a local minimizer do not reduce to a singleton as $\ell \to \infty$.

Basic theorem

Theorem

Assume that ***A.0*****–*****A.3*** *hold. Then, for almost every sampling process* $\bar{\xi}$*,* $\left\{z^* \in \mathcal{Z}\left(\{z_N^*(\bar{\xi})\}\right)\right.$ *satisfying* ***A.4***$\left.\right\}$ *is a set of local minimizers of* $g(\cdot)$ *over* S*.*

Proof.

See Bastin, Cirillo, and Toint (2006), *Convergence theory for nonconvex stochastic programming with an application to mixed logit*, Mathematical Programming, 108(2–3):207–234. □

Stochastic constraints

Assumption **A.4** is somewhat artificial and it is therefore more interesting to look for more elegant conditions.

Unlike classical sensitivity analysis, we focus here on conditions under which a limit point of a sequence of approximate solutions is a solution of the true problem.

As previously, we consider the case where the feasible set is described by a set of equality and inequality constraints, and we assume that ϵ_n converges to 0 almost surely, uniformly on V.

Lagrange multipliers convergence

Consider a particular sampling process $\overline{\xi}$ in $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$, and $z^* \in \mathcal{Z}\left(\{z_n^*(\overline{\xi})\}\right)$. Under some conditions, if the subsequence $\{z_\ell^*\}_{\ell=1}^\infty \to z^*$, then $\{\lambda_\ell^*\} \to \lambda^*$, associated to z^*.

Lemma

Consider a particular sampling process $\overline{\xi}$ in $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$ such that $\epsilon_n\left(z, \overline{\xi}\right) \to 0$ uniformly on V as $n \to \infty$. If $z^* \in \mathcal{Z}\left(\{z_n^*(\overline{\xi})\}\right) \cap \overset{o}{V}$, $\{z_\ell^*\}_{\ell=1}^\infty \subseteq \{z_n^*\}_{n=1}^\infty \cap \overset{o}{V}$ converges to z^*, and there exists some unique Lagrange multipliers vector associated to z^* satisfying the KKT conditions, then $\lambda_\ell^*(\overline{\xi})$ converges to λ^* as $\ell \to \infty$.

LICQ

The unicity of λ^* can be ensured with some appropriate constraints qualification condition, as the linear independence constraint qualification (LICQ).

Definition

Consider z^* and the active set $\mathcal{A}(z^*)$:

$$\mathcal{A}(z^*) = \{i \in \{1, \ldots, k\} \mid c_i(z^*) = 0\} \cup \{k+1, \ldots, M\}.$$

The LICQ holds at z^* if the set of active constraints gradients $\{\nabla c_j(z^*),\ j \in \mathcal{A}(z^*)\}$ is linearly independent.

Strict complementarity

Another useful concept for our needs is the strict complementarity condition.

Definition

Given z^* and a vector λ^* satisfying the KKT conditions, the strict complementarity condition holds if $[\lambda]_j^* > 0$ for each $j \in \{1, \ldots, k\} \cap \mathcal{A}(z^*)$.

Consider again a particular sampling process $\overline{\xi}$ in $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$, such that $\epsilon_n(z, \overline{\xi}) \to 0$ as $n \to \infty$. If the assumptions of the above lemma hold for some subsequence $\{z_\ell^*\}_{\ell=1}^\infty \to z^*$, the gradient of the Lagrangian $\nabla L\left(z_\ell^*(\overline{\xi}), \lambda_\ell^*(\overline{\xi})\right)$ converges to $\nabla \mathcal{L}(z^*, \lambda^*)$, as ℓ tends to infinity, with $\lambda_\ell^*(\overline{\xi}) \to \lambda^*$.

Strict complementarity (cont'd)

Assume that the strict complementarity condition holds at z^* for the approximate problem. Then, for ℓ large enough,

$$[\lambda_\ell^*]_j(\bar{\xi}) > 0, \ \forall j \in \{1, \ldots, k\} \cap \mathcal{A}(z^*).$$

Thus the corresponding constraints are active at $z_\ell^*(\bar{\xi})$.

Moreover, as $\epsilon_n(z, \bar{\xi}) \to 0$, $\hat{c}_j\left(z_\ell^*(\bar{\xi}), \epsilon_\ell(z_\ell^*(\bar{\xi}), \bar{\xi})\right) \to c_j(z^*)$ and, for ℓ large enough, $\mathcal{A}\left(z_\ell^*(\bar{\xi})\right) = \mathcal{A}(z^*)$, so the strict complementarity condition holds at $\left(z_\ell^*(\bar{\xi}), \lambda_\ell^*(\bar{\xi})\right)$ for the approximate problem.

A second-order convergence result

Theorem (Second-order convergence)

Assume that, for almost every sampling process $\overline{\xi}$ *in* $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$*, there exists some* $z^* \in \mathcal{Z}\left(\{z_n^*(\overline{\xi})\}\right) \cap \overset{o}{V}$ *associated to a unique vector of Lagrange multipliers* λ^* *and some subsequence* $\{z_\ell^*\}_{\ell=1}^\infty \subseteq \{z_n^*\}_{n=1}^\infty$*, such that* $z_\ell^*(\overline{\xi}) \to z^*$*, and*

(a) $\epsilon_n\left(z_n^*(\overline{\xi}), \overline{\xi}\right) \to 0$ *uniformly on* V*, as* $n \to \infty$*,*

(b) $z_n^*(\overline{\xi}) \in \overset{o}{V}$*,* $n = 1, \ldots,$

(c) $\nabla_{zz}^2 \hat{g}\left(z_\ell^*(\overline{\xi}), \epsilon_\ell(z_\ell^*(\overline{\xi}), \overline{\xi})\right) \to \nabla_{zz}^2 g(z^*)$ *as* $\ell \to \infty$*,*

(d) $\nabla_{zz}^2 \hat{c}_j\left(z_\ell^*(\overline{\xi}), \epsilon_\ell(z_\ell^*(\overline{\xi}), \overline{\xi})\right) \to \nabla_{zz}^2 c_j(z^*)$ $(j = 1, \ldots, M)$ *as* $\ell \to \infty$*.*

We also assume that the strict complementarity condition and the LICQ hold at (z^*, λ^*) *for the true problem.*

A second-order convergence result (cont'd)

Theorem (Second-order convergence (cont'd))

Then, for almost every sampling process $\bar{\xi}$*,*

(i) *the LICQ holds at* $z_\ell^*(\bar{\xi})$*, for* ℓ *large enough,*

(ii) (z^*, λ^*) *satisfies the second-order necessary condition for the true program:*

$$w^T \nabla^2_{zz} \mathcal{L}(z^*, \lambda^*) w \geq 0, \text{ for all } w \in \mathit{Null} \left[\nabla_z c_j(z^*)^T \right]_{j \in \mathcal{A}(z^*)}.$$

If moreover $\exists$ *some constant* $\alpha_{\bar{\xi}} > 0$ *such that,* $\forall$ ℓ *large enough,*

$$w^T \nabla^2_{zz} L_\ell \left(z_\ell^*(\bar{\xi}), \lambda_\ell^*(\bar{\xi}) \right) w > \alpha_{\bar{\xi}},$$
$$\text{for all } w \in \mathit{Null} \left[\nabla_z \hat{c}_j(z_\ell^* \left(\bar{\xi} \right))^T \right]_{j \in \mathcal{A}(z_\ell^*(\bar{\xi}))}, \|w\| = 1,$$

then (z^*, λ^*) *almost surely satisfies the sufficient conditions for the true problem.*

A second-order convergence result (cont'd)

Theorem (Second-order convergence (cont'd))

In other words

(iii) $w^T \nabla^2_{zz} \mathcal{L}(z^*, \lambda^*) w > 0,$

$$\text{for all } w \in \text{Null}\left[\nabla_z c_j(z^*)^T\right]_{j \in \mathcal{A}(z^*)}, \|w\| = 1.$$

This is equivalent to saying that z^* is an isolated minimizer for the original problem.

Proof.

See Bastin, Cirillo and Toint, 2006. □

Discussion

Remark that the LICQ and the strict complementarity condition imply that the minimizer is isolated while sufficient second-order conditions are usually used to characterize strict local minimizers.

If $z^* \in \mathcal{Z}\left(\{z_n^*(\bar{\xi})\}\right) \cap \overset{o}{V}$ is a strict but not isolated local minimizer, every neighborhood of z^* contains other local minimizers that are candidates to be limit points of sequences of SAA solutions, as n grows to infinity, and z^* can therefore be difficult to identify.

Non-degeneracy

The non-degeneracy assumption

$$w^T \nabla^2_{zz} L_\ell \left(z^*_\ell(\bar{\xi}), \lambda^*_\ell(\bar{\xi})\right) w > \alpha_{\bar{\xi}}, \quad \text{for all } w \in \text{Null}\left[\nabla_z \hat{c}_j(z^*_\ell\left(\bar{\xi}\right))^T\right]_{j \in \mathcal{A}(z^*_\ell(\bar{\xi}))}, \|w\| = 1,$$

can also be replaced by requiring that the Jacobian of equality equations involved in the KKT conditions associated to the original program

$$\begin{aligned} \nabla_z \mathcal{L}(z, \lambda) &= 0, \\ [\lambda]_j c_j(z) &= 0, \ j = 1, \ldots, M, \\ c_j(z) &= 0, \ j = k+1, \ldots, M \end{aligned}$$

is non-singular at (z^*, λ^*).

Second-order convergence: corollary

Corollary

Assume that, for almost every sampling process $\bar{\xi}$ *in* $(\Xi_\Pi, \mathcal{F}_\Pi, P_\Pi)$*, there exist some* $z^* \in \mathcal{Z}\left(\{z_n^*(\bar{\xi})\}\right) \cap \overset{o}{V}$ *associated to a unique Lagrange multipliers vector* λ^* *and some subsequence* $\{z_\ell^*\}_{\ell=1}^\infty \subseteq \{z_n^*\}_{n=1}^\infty$*, such that* $z_\ell^*(\bar{\xi}) \to z^*$*, that assumptions (a)–(d) of the second-order convergence theorem hold and that the strict complementarity condition holds at* (z^*, λ^*) *for the original program.*

Assume moreover that the Jacobian of the equality relations of the KKT conditions is non-singular at (z^*, λ^*)*. Then there exists almost surely some* z^* *in* $\mathcal{Z}\left(\{z_N^*(\bar{\boldsymbol{\xi}})\}\right) \cap \overset{o}{V}$*, associated to some vector* λ^**, that satisfies the sufficient second-order conditions of the original program.*

Overfitting

Consider again a deterministic feasible set.
Let $z^* \in \arg\min g(z)$ and $z_N^* \in \arg\min \hat{g}_N(z)$ for a given sample $\{\xi_1, \xi_2, \ldots, \xi_N\}$.
We have $\forall y \in S$

$$\hat{g}_N(y) \geq \hat{g}_N(z_N^*) = \min_{z \in S} \hat{g}_N(z)$$

Taking the expectation on both sides, we obtain

$$g(y) = \mathbb{E}[\hat{g}_N(y)] \geq \mathbb{E}\left[\min_{z \in S} \hat{g}_N(z)\right]$$

Since this is true for any y, we have in particular

$$\mathbb{E}\left[\min_{z \in S} \hat{g}_N(z)\right] \leq g(z^*) = \min_{z \in S} g(z).$$

In other words, for N finite, the optimal value estimator is biased, but asymptotically unbiased.

Out-of-sample validation

Choose $M >> N$, and compute $\hat{g}_M(z_N^*)$. We hope $\hat{g}_M(z_N^*) \approxeq g(z_N^*)$.

Given N, we can estimate $\mathbb{E}[\min_{z \in S} \hat{g}_N(z)]$ by averaging over J replications:

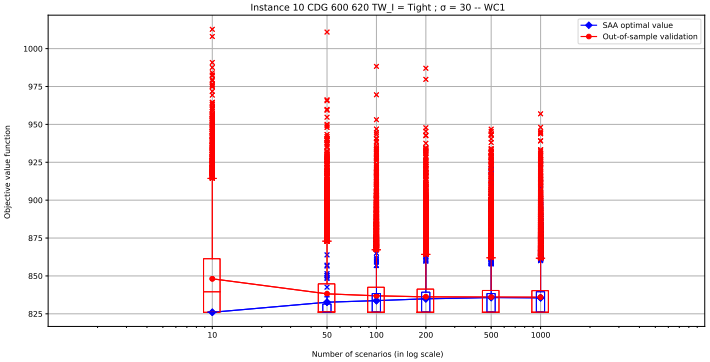
$$\mathbb{E}\left[\min_{z \in S} \hat{g}_N(z)\right] \approxeq \frac{1}{J} \sum_{j=1}^{J} \min_{z \in S} \hat{g}_N\left(z, \xi_N^{(j)}\right).$$

We expect

$$\frac{1}{J} \sum_{j=1}^{J} \min_{z \in S} \hat{g}_N\left(z, \xi_N^{(j)}\right) \leq \frac{1}{J} \sum_{j=1}^{J} \hat{g}_M\left(z_N^{*,(j)}\right),$$

with a decreasing difference as N grows.

Out-of-sample validation: example



Ahmed Khassiba (2020), "Two-stage stochastic programming for aircraft arrival scheduling under uncertainty", Ph.D. thesis.

SAA - Chance constraints

We consider the problem

$$\min_{x \in \mathcal{X}} f(x)$$
$$\text{s.t. } P[G(x, \xi) \leq 0] \geq 1 - \alpha,$$

where $G : \mathcal{R}^n \times \Xi \to \mathcal{R}^m$.
This allows to consider separate and joint chanced constrained.
We can reformulate this problem as

$$\min_{x \in \mathcal{X}} f(x)$$
$$\text{s.t. } P[G(x, \xi) > 0] \leq \alpha,$$

If $P[G(x, \xi) \leq 0]$ has no closed form, we can always approximate it by sample average approximation.

SAA

With the indicator function of $(0, \infty)$

$$\mathcal{I}_{(0,\infty)} = \begin{cases} 1 & \text{if } t > 0, \\ 0 & \text{if } t \leq 0, \end{cases}$$

we can rewrite the chance constraint as

$$p(x) = \mathbb{E}\left[\mathcal{I}_{(0,\infty)}\left(G(x,\xi)\right)\right] \leq \alpha.$$

Let $\xi_1, \ldots, \xi_N$ be an i.i.d. sample of N realizations of ξ. We approximate the chance constraint by

$$\hat{p}_N = \frac{1}{N}\sum_{j=1}^{N} \mathcal{I}_{(0,\infty)}\left(G(x,\xi_j)\right) \leq \gamma$$

SAA (cont'd)

$\hat{p}_N(x)$ is equal to the proportion of times that $G(x, \xi_j) > 0$. γ is not necessarily equal to α, especially when we want to enlarge the feasible of the approximate problem.

The approximate problem is

$$\begin{aligned} &\min_{x \in \mathcal{X}} f(x) \\ &\text{s.t. } \hat{p}_N(x) \leq \gamma. \end{aligned}$$

Theory

For theoretical justifications, see

B.K. Pagnoncelli, S. Ahmed, and A. Shapiro (2009), "Sample Average Approximation Method for Chance Constrained Programming: Theory and Applications", Journal of Optimization Theory and Applications 142 (2), pp. 399–416.

Especially, under some regularity conditions, we have

$$sup_{x \in C} |\hat{p}_N(x) - p(x)| \to 0,$$

w.p.1, as $N \to \infty$.

Convergence

Assume that there is an optimal solution $\overline{x}$ of the true problem such that for any $\epsilon > 0$ there is $x \in \mathcal{X}$ with $\|x - \overline{x}\| \leq \epsilon$ and $p(x) < \alpha$.

Theorem

Suppose $\gamma = \alpha$, $\mathcal{X}$ compact, and $f(x)$ continuous. Then $\hat{v}_N \to v^*$, where $\hat{v}_N$ and v^* are the optimal values of the approximate and true problems, respectively, and $D(\hat{S}_N, S) \to 0$ almost surely as $N \to \infty$.

Application: call centers

Joint work with Thuy Anh Ta, Tien Mai, Wyean Chan, and Pierre L'Ecuyer. The next slides are due to Thuy Anh Ta.

A call center: an office used for receiving or transmitting customers requests by telephone.

- A call represents a customer and it is classified by type (K types).
- Customers are served by agents.
- Agents sharing a common set of tasks form a group of agents (I groups).
- A day is divided into periods (P periods).
- The work schedule of an agent is determined by her shift (Q shift configurations).

Performance measure

- Performance measures (PM) allow to assess the quality of service (QoS) and efficiency of a call center, e.g., Service level (SL) and average waiting time (AWT), abandonment ratio, occupancy ratio, etc.
- Service level (SL): the percentage of calls that are answered in a defined waiting threshold.
- Average waiting time (AWT): the average (or mean) time a customer waited to have a service.
- QoS can be defined in the long run (expectations), or over a given time period (random variable).

Staffing, Scheduling and Routing

- The objective is to minimize the operating cost of a call center under a set of constraints on certain PMs (SL or AWT).
- Staffing: deciding the number of agents $y_{i,p}$ of each group for each period in a day.
- Scheduling: determining how many agents $x_{i,q}$ to assign to a set of predefined shifts.
- Routing: determining which agents are allowed to handle the call, and how agents are chosen when several agents are free.

Modeling a call center

- The calls arrive according to arbitrary stochastic processes that could be non-stationary (e.g. homogeneous Poisson process), and perhaps doubly stochastic (e.g. Poisson gamma process, the arrival rate is a random variable of gamma distribution).
- Service times: exponential, log-normal distributions.
- Delay time, patience time, etc.
- When number of skills is low: Arrivals are served in a first come first serve (FCFS) order and/or a longest idle server first rules.
- Otherwise, skills based routing may be used to route calls to appropriate agents.

Staffing and scheduling with predictable arrival rates

$$g(y) = \frac{\mathbb{E}[\#\text{of served call that waited at most } \tau]}{\mathbb{E}[\#\text{ Total } \#\text{ of calls}]}$$

Scheduling problem

$$\min c^T x = \sum_{i=1}^{I} \sum_{q=1}^{Q} c_{i,q} x_{i,q}$$

subject to:

$$\begin{aligned} Ax &\geq y \\ g_{k,p}(y) &\geq l_{k,p} \ \forall k, p, \\ g_p(y) &\geq l_p \ \forall p, \\ g_k(y) &\geq l_k \ \forall k, \\ g(y) &\geq l, \\ x, y &\geq 0 \text{ and integer.} \end{aligned} \tag{P1}$$

Staffing and scheduling with predictable arrival rates

Staffing problem

$$\min_{y} c'^{T} y = \sum_{i=1}^{I} \sum_{p=1}^{P} c'_{i,p} y_{i,p}$$

subject to:

$$\begin{aligned} g_{k,p}(y) &\geq l_{k,p} \ \forall k, p, \\ g_p(y) &\geq l_p \ \forall p, \\ g_k(y) &\geq l_k \ \forall k, \\ g(y) &\geq l, \\ y &\geq 0 \text{ and integer.} \end{aligned} \tag{P2}$$

Staffing and scheduling

SL $S(\tau, y)$ and AWT $W(y)$ are considered as r.v. in a given period.
The chance-constrained scheduling problem for multiskill call centers is formulated as:

$$\min c^T x = \sum_{i=1}^{I} \sum_{q=1}^{Q} c_{i,q} x_{i,q}$$

subject to:

$$\begin{aligned} Ax &\geq y \\ \mathbb{P}[S_{k,p}(\tau_{k,p}, y) \geq s_{k,p}] &\geq r_{k,p} \qquad \forall k, p \\ \mathbb{P}[W_{k,p}(y) \leq w_{k,p}] &\geq v_{k,p} \qquad \forall k, p \\ x, y &\geq 0 \text{ and integer.} \end{aligned} \tag{P3}$$

Sample average approximate problem

$$\min c^T x = \sum_{i=1}^{I} \sum_{q=1}^{Q} c_{i,q} x_{i,q}$$

subject to:

$$\begin{array}{lll} Ax & \geq y & \\ \frac{1}{N} \sum_{d=1}^{N} \mathbb{I}[\hat{S}^d_{N,k,p}(\tau_{k,p}, y) \geq s_{k,p}] & \geq r_{k,p} & \forall k, p \\ \frac{1}{N} \sum_{d=1}^{N} \mathbb{I}[\hat{W}^d_{N,k,p}(y) \leq w_{k,p}] & \geq v_{k,p} & \forall k, p \\ x, y & \geq 0 \text{ and integer} . & \end{array} \qquad \text{(SAA)}$$

Under some specific conditions, the optimal value and the set of optimal solutions of the SAA problem converge to those of the true problem with probability one as N approaches infinity.